Escape hatches are phase edges, but not all phase edges are escape hatches*

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1. Introduction

Syntactic movement between locality domains is known to be restricted, which in turn necessitates specific strategies to allow for long-distance dependencies. Among these strategies are pied-piping, i.e., moving the containing domain itself to bring a contained element closer to its intended position, or successive cyclic movement (SCM), where the moving element is allowed to move out of a locality domain through a non-complement position (or *edge*). Minimalist syntax standardly characterizes such domains as *phases* (Chomsky 2000, 2001). As proposed by Chomsky (2000), SCM is required because chunks of syntactic structure are periodically *transferred* to the interfaces, rendering them unavailable to higher operations. By stipulation, Transfer targets the complement of a phasal category, leaving its edge available as a kind of *escape hatch* for further extraction.

However, the conceptual motivations and precise mechanisms of phase theory remain unsettled. Standard phasal categories (C, *v*, and often D and P) do not form an uncontroversial natural class, and the exemption of edges from Transfer remains largely stipulative. But a more fundamental issue is how the empirical picture is interpreted theoretically to begin with. The empirical landscape indicates that if an element moves out of a phase[1], it must stop at its edge. Phase theory often interprets this to mean the edge itself *must* be transparent, whereas the extraction data only demonstrate that it *may* be.

This paper presents new evidence that the edges of certain phasal categories (viz. C and D) can be opaque (i.e. some phases are effectively islands), challenging the standard assumption that edges are always transparent. Section 2 demonstrates this opacity in the nominal domain, drawing on microvariation in Dutch *what-for*-constructions and Split Topicalization, alongside a Malayalam adnominal fronting construction. Section 2 also estab-

*We would like to thank the audience at NELS 56 and Hedde Zeijlstra for feedback on this project. Work on this paper was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) - GRK2636, whose support is gratefully acknowledged.

[1]Or more accurately, it moves out of something identifiable as a phase within a given grammar. It may be that the categories that are phase(-like) are not the same across languages, particularly since the inventories of syntactic categories need not otherwise be the same across languages.

lishes parallel effects in the clausal domain in Mongolian A-dependencies and Hungarian A'-dependencies. We conclude on the basis of this typological variation that phase edge opacity is the grammatical default, with escape hatch effects arising only exceptionally.

We sketch a revision to phase theory on this basis which accommodates this apparent variability in edge opacity. In Section 3, we show how the distinction between opaque and transparent edges may be derived from the relative ordering of Labelling and Transfer. While the rigid Phase Impenetrability Condition (PIC) struggles to capture this cross-linguistic variation, our approach better accounts for the empirical landscape and provides a principled conceptual grounding for escape hatches. We therefore argue for deprecating the strict PIC in favour of this flexible alternative, and conclude by assessing the broader theoretical consequences in Section 4.

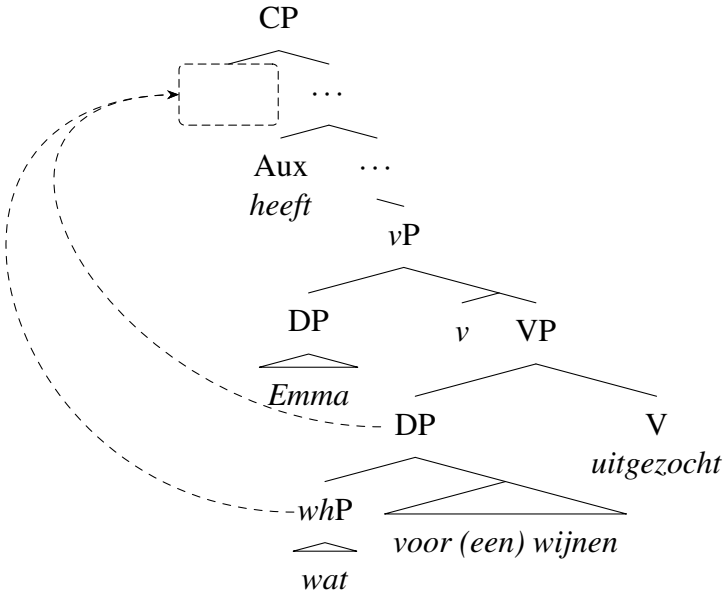
2. DP data

The first piece of empirical evidence of some phasal edges actually being opaque comes from *what-for*-constructions (WFCs) and Split Topicalization (ST) in Dutch. Both phenomena involve the movement of syntactic material from inside the DP to a position in the left periphery. Focusing on WFCs, the derivation involves movement of a *wh*-element. This *wh*-element either pied-pipes the entire DP (1a), or it splits off of the DP it originates from (1b), stranding the rest of the DP. The former case will be referred to as the non-split variant, and the latter as the split variant. The two possibilities are shown in the assumed structure in (2).

(1) a. [Wat voor (een) wijnen]$_i$ heeft Emma t_i uitgezocht
What for a wines has E. chosen
'What kind of wines did Emma choose?' (Dutch non-split WFC)

b. Wat$_i$ heeft Emma [t_i voor (een) wijnen] uitgezocht
What has E. for a wines chosen
'What kind of wines did Emma choose?' (Dutch split WFC)

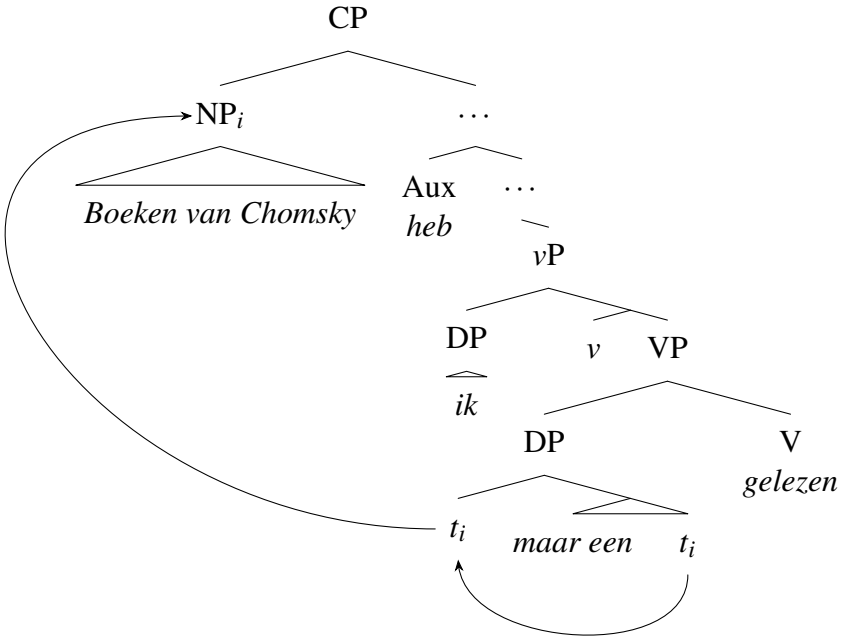
(2)



ST is subject to a similar process, albeit different in the syntactic material that undergoes movement. In ST, the derivation involves movement of an NP out of a DP into the left periphery (3), leaving the remnant of the DP in the VP, as shown in (4).

(3) [Boeken van Chomsky]$_i$ heb ik maar [een t$_i$] gelezen
Books of Chomsky have I but one read
'As for books by Chomsky, I have read only one.' (Dutch ST)

(4)



Given their structural parallels, microvariation in the acceptability of WFCs (1) and ST (3) across Dutch dialects provides crucial insight into the underlying grammar. A recent survey of 277 dialect speakers by Hendriks (To appear) identifies three distinct acceptability profiles regarding non-split WFCs, split WFCs, and ST: Group A (N=32) strictly allows non-split WFCs; Group B (N=188) accepts both split and non-split WFCs; and Group C (N=56) permits both WFC variants alongside ST. An overview is given in (5).

(5) Acceptability of non-split WFCs, split WFCs and ST by Group

	Non-split WFCs	Split WFCs	ST
Dutch Group A	Yes	No	No
Dutch Group B	Yes	Yes	No
Dutch Group C	Yes	Yes	Yes

Assuming split WFCs and ST involve A'-subextraction of *wh*-elements and NPs respectively, these data yield two principal conclusions. First, the availability of A'-subextraction from DPs is subject to inter-speaker variation. Group A's rejection of both constructions indicates absolute extraction opacity, directly challenging the assumption of uniformly transparent phase edges. Furthermore, the optionality observed in Groups B and C, who accept both split and non-split WFCs, demonstrates that reaching the DP edge does not necessitate further extraction. Instead, subsequent movement is presumably driven by discourse-structural properties.[2]

Second, extraction is sensitive to the categorial status of the moved element. While a majority accepts split WFCs, only a minority (Group C) tolerates ST (see also Hendriks 2023). Assuming that the derivational divide between these constructions reduces to the extractee's category, DP edge transparency is demonstrably selective: for most Dutch speakers, the DP edge is transparent to *wh*-elements, but remains strictly opaque to NPs.

2.1 Malayalam data

Malayalam (Dravidian) exhibits a construction whose grammar parallels that of Dutch Group A, which only accepts non-split WFC. However, instead of *wh*-movement, this construction involves movement of an adnominal adjunct[3] to the edge of NP/DP, but not out of it. We will call this construction the adnominal fronting construction (AdFr).

As background, Malayalam allows for arguments to be displaced somewhat freely, including to the left edge of the clause as shown in (6).

(6) a. Sīta Rāmaṉ-e kaṇṭu
S. R.-ACC see.PST
'Sita saw Raman.'

b. Rāmaṉ-e$_i$ Sīta ____$_i$ kaṇṭu
R.-ACC S. see.PST
'Raman, Sita saw.'

[2]For theoretical derivations of standard Topicalization, see Platzack (1996) or Craenenbroeck (2006).

[3]Malayalam has very few (if any) "true" adjectives. Adnominal adjuncts in Malayalam almost always end in *-a*. See Menon (2016) for details.

Perhaps unsurprisingly, the same applies to adjuncts in the clause, as in (7):

(7) a. Rāmaṉ vēgam vannu
R. quickly come.PST
'Raman quickly came.'

b. Vēgam$_i$ Rāmaṉ ——$_i$ vannu
quickly R. come.PST
'Quickly, Raman came.'

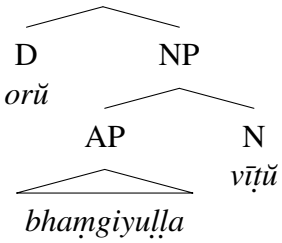
In the nominal domain, Malayalam, like English, has the default order D $\prec$ (Num) $\prec$ A $\prec$ N. The language does not have a definite article, but this order can be seen with the indefinite article *orŭ* (lit. 'one') in (8a). In contrast, the AdFr construction exhibits marked alternative order where the adnominal modifier may precede the article (8b).

(8) a. Rāmaṉ [orŭ bhaṃgiyuḷḷa vīṭŭ] vaṅṅi
R. INDEF beautiful house buy.PST
'Raman bought a beautiful house.'

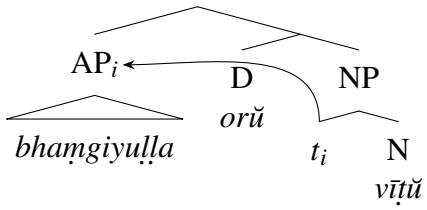
b. Rāmaṉ [bhaṃgiyuḷḷa$_i$ orŭ ——$_i$ vīṭŭ] vaṅṅi
R. beautiful INDEF house buy.PST
'Raman bought a *beautiful* house.' (Malayalam AdFr)

We can reasonably guess that this is a kind of A'-movement, as it simply triggers a somewhat marked information-structural effect (similar to scrambling phenomena, though the exact effect is immaterial here). Assuming that the indefinite article *orŭ* here projects a DP[4], we expect the structures in (9) for the nominals in question in (8):

(9) a.



b.



If NPs/DPs are phasal à la Bošković (2005, *et seq.*), we again expect that its edge is exempt from Transfer, and as such, the AP should be visible to outside structure. Since we know that A'-movement to the left edge of CP is possible, and that A'-movement to the left edge of NP/DP has occurred in this case, Malayalam may be expected to allow left-branch extraction (LBE) of the adnominal from the nominal domain, especially since, like languages with LBE, it has no definite article. This is not borne out — LBE of these adnominals is globally disallowed in Malayalam[5]:

[4]One may object that it is not necessary that *orŭ* constitutes a D head, but may instead be a dependent of the NP. Even if this is the case, it does not affect the overall argument above.

[5]A valid question is whether this movement fails because of the category of the adnominal rather than any phase-related issues. However, considering that various different syntactic categories may be displaced across a distance within the clausal domain in Malayalam, there is no obvious reason for this to be the case.

(10) *Bhaṃgiyuḷḷa$_i$ Rāmaṉ [____$_i$ orŭ ____$_i$ vīṭŭ] vaṅṅi
beautiful R. INDEF house buy.PST
'Raman bought a *beautiful* house.' (Malayalam failed LBE)

One interpretation of this is that Malayalam DPs have a grammar that is parallel to that of Dutch speakers in Group A, who disallow split WFCs: despite the presence of an otherwise mobile element at the left edge of NP/DP, further movement out of NP/DP is blocked. This confirms that the behaviour observed for Group A in the Dutch study is not exceptional, but rather something that can be observed in NPs/DPs across languages.

2.2 Interim summary

Assuming that the nominal domain (NP/DP) is phasal, on standard assumptions, we expect uniform transparency of its left edge. However, microvariation in Dutch dialects in WFCs and ST, and Malayalam AdFr are rather unexpected: all speakers of Dutch should accept split WFCs and ST, and LBE should be possible of any Malayalam elements which undergo AdFr[6]. However, as was shown in this section, this is not what we find. The reader may wonder if this is a situation unique to DPs; however, we will show in the next section that in the case of CPs, which are considered one of the canonical environments in which escape hatches occur, similar cases of edge opacity may be found. This, we will argue, motivates a revisiting of the underlying assumptions of phase theory.

2.3 CP data

As foreshadowed by the DP data, movement to a phase edge does not entail subsequent extractability. This holds true for the CP domain as well. For example, Horvath (1997) demonstrates that alongside standard SCM through CP (11a), Hungarian permits partial *wh*-movement: the *wh*-phrase stops at the embedded Spec,CP, triggering the insertion of a *wh*-expletive (WHE) in the matrix Spec,CP (11b).

(11) a. Kivel$_i$ akarod t$_i$ hogy beszéljek
with_whom want.2SG that speak.SBJV.1SG
'With whom do you want me to talk?'
(Hungarian long-distance movement; Horvath 1997:533)

b. Mit gondolsz, hogy kit látott János
what think.2SG that who.ACC see.PST.3SG John.NOM
'Who do you think that John saw?'
(Hungarian WHE; Horvath 1997:510)

Crucially, Horvath shows that the WHE is always used when its associated *wh*-phrase is within an island. This is reminiscent of variably splittable WFCs in Dutch Group B/C.

[6]This of course says nothing of factors external to the syntax; we do not focus here on the information structure, which is known to play a crucial role.

An explanation for the optionality in (11) is that there are two derivations: one with a transparent edge and SCM, and one with an opaque one, where the WHE is inserted as a repair.

An even clearer piece of empirical evidence for variably opaque edges comes from Mongolian. Mongolian embedded subject variants (ESVs) can either be marked accusative, trigger Condition B effects, and have the option of hyperraising (12a), or they can be marked nominative and remain in the embedded CP (12b).

(12) a. Odgerel$_i$ [margaash tüün-iig$_{*i/j}$ ir-ne gej] khel-sen
O. tomorrow 3SG-ACC come-N.PST COMP say-PST
'Odgerel$_i$ said that (s)he$_j$ is coming tomorrow.'

b. Odgerel$_i$ [margaash ter$_{i/j}$ ir-ne gej] khel-sen
O. tomorrow 3SG.NOM come-N.PST COMP say-PST
'Odgerel$_i$ said that (s)he$_{i/j}$ is coming tomorrow.'

(Mongolian ESVs; Fong 2019:13)

Fong (2019) interprets this as evidence that embedded subjects can optionally move to embedded Spec,CP, or remain in Spec,TP. If our intuitions are correct, a simpler explanation is that there is a fixed position for ESVs which is either an opaque or transparent phase edge. If it is transparent, the embedded subject may be accessible to binding, case assignment, movement, etc. whereas if it is opaque, none of those properties obtain. This is again parallel to the variable splittability of WFCs for some Dutch speakers.

2.4 Summary of comparison between DP and CP data

The DP phenomena from Dutch and Malayalam and the CP phenomena from Hungarian and Mongolian that have been discussed can be divided into two categories: one where the phase edge is transparent and another where it is opaque. An overview of the empirical picture that has been sketched thus far is summarized in (13).

(13) Distribution of edge opacity by category

	Edge opaque	Edge transparent
DP	Dutch non-split WFCs; Malayalam AdFr	Dutch split WFCs; Dutch ST
CP	Hungarian WHE; Mongolian NOM ESVs	Hungarian SCM; Mongolian ACC ESVs

The cross-linguistic observations summarized in the table above lead to a one-way implication: moving outside a phase requires being at an edge, but being an edge does not automatically allow escaping it. In other words, edges sometimes seem to be opaque. This raises the question of how the variability in the status of phase edges (i.e., transparent versus opaque) should be encoded in the grammar. In the next section, we will sketch one possible approach that can capture the logical space (13), to illustrate the kind of work needed to bring phase theory closer to the empirical picture.

3. Proposal

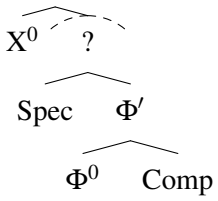
To derive the observed cross-linguistic variability in phase edge transparency, we propose that edge opacification reduces to the relative ordering of Transfer and Labelling. We assume Transfer targets the internal structure of a set formed by Merge without removing the set itself. Hence, the contents of a transferred set $\{\alpha, \beta\}$ become syntactically opaque while its label remains visible, i.e. $\{_{\alpha}\ \cancel{\alpha, \beta}\}$. Labelling is only possible before the internal structure of a set is transferred, and as selection requires a label, a set must be labeled before its maximal projection undergoes Transfer; otherwise, it is unselectable.

We hypothesize that a derivation involves competing instructions which may be ordered in an optimal way, such that the derivation converges. On this basis, we conceptualize Transfer and Labelling as two such derivational instructions which are in competition. We posit that phasal categories like C and D bear a $\textsc{Tr}_{\Phi}$ instruction, which tells them to transfer as much internal structure as possible (including the head, cf. Boeckx and Grohmann 2004). Conversely, when a specifier merges, a separate instruction $\textsc{Lb}_{\text{Spec}}$ must be followed, which instructs the grammar to label the newly formed syntactic object.

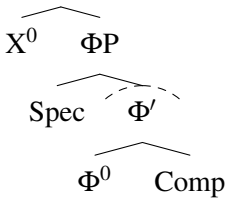
The empirical variation emerges during the derivation of the phasal phrase. A phase head Φ^0 merges with its complement and projects: $\{_{\Phi'}\ \Phi, \text{Comp}\}$. Merging the specifier thereafter yields $\{\text{Spec}, \{_{\Phi'}\ \Phi, \text{Comp}\}\}$, which requires both $\textsc{Tr}_{\Phi}$ and $\textsc{Lb}_{\text{Spec}}$. The resulting edge transparency strictly depends on which instruction is followed first.

If Transfer applies first, targeting the maximal unlabelled set $\{\cancel{\text{Spec}, \{_{\Phi'}\ \Phi, \text{Comp}\}}\}$ would render the phase unselectable (14a). The optimal derivation that avoids a crash then only transfers the previously labeled intermediate projection: $\{\text{Spec}, \cancel{\{_{\Phi}\ \Phi, \text{Comp}\}}\}$. Subsequent application of $\textsc{Lb}_{\text{Spec}}$ labels the entire phrase, yielding $\{_{\Phi\text{P}}\ \text{Spec}, \cancel{\{_{\Phi'}\ \Phi, \text{Comp}\}}\}$. Thus, the specifier survives Transfer, deriving the standard escape hatch effect (14b). If Labelling applies first, the complete phase successfully projects: $\{_{\Phi\text{P}}\ \text{Spec}, \{_{\Phi'}\ \Phi, \text{Comp}\}\}$. With the label available, the phrase is selectable, and following $\textsc{Tr}_{\Phi}$, the whole internal structure of the set may be transferred: $\{_{\Phi\text{P}}\ \cancel{\text{Spec}, \{_{\Phi'}\ \Phi, \text{Comp}\}}\}$. This includes the specifier, effectively neutralizing the escape hatch and yielding a kind of island (14c).

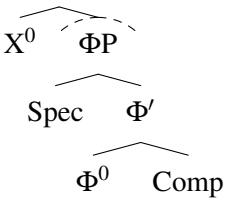
(14) a.



b.



c.



The order in which these instructions are followed thus determines whether a specifier is transparent or opaque. By default, we assume that the grammar prioritizes Labelling, meaning phase edges are generally opaque. In Malayalam AdFr, for example, Labelling is prioritized over Transfer, which leads to the DP being labeled earlier, which causes the specifier to become opaque for subextraction. The same can be said for Dutch grammars that disallow split WFCs (i.e., Dutch Group A). To capture the variability, then, we must

assume exceptions to the default ordering. In Dutch ST, for instance, the inverse holds: the instruction to transfer is prioritized over the instruction to label, preventing the specifier from being transferred with the rest of the phase, which in turn allows subextraction[7].

In some cases, however, no priority seems to hold for the ordering of the two instructions, causing the grammar to generate two options. As such, variability within constructions/phenomena can be observed. This is the case in Dutch WFCs (in Groups B and C), Hungarian WHE constructions and Mongolian ESVs. When the default ordering ($\text{LB}_{\text{Spec}} > \text{TR}_{\Phi}$) is in place, the edges are opaque, meaning non-split WFCs, WHE constructions and nominative ESVs are generated. With the alternative order ($\text{TR}_{\Phi} > \text{LB}_{\text{Spec}}$) the edges are transparent, hence allowing split WFCs, non-WHE constructions (i.e., regular SCM) and accusative ESVs. Which of the two derivations surfaces is of course filtered by downstream factors like information-structure.

Finally, in addition to the ordering of Transfer and Labelling, the empirical data tell us that the syntactic identity of the specifier of a phase matters as well. Recall that the difference between Dutch Group B and Dutch Group C concerned the acceptability of ST. Both ST and split WFCs require Transfer to be prioritized over Labelling. However, if that is all a grammar requires to generate the syntactic constructions, then either both ST and split WFCs would be acceptable, or neither. The observed distinction between the two constructions is therefore attributed to the syntactic category of the material that resides in Spec,DP. To account for the variability, we suggest that the ordering of the two instructions must be dependent on the type of specifier involved to some extent. That is, some Dutch grammars allow Transfer to be prioritized over Labelling only if the specifier involved is a *wh*P (i.e., Dutch Group B), while other grammars allow this when the specifier is a *wh*P or NP (i.e., Dutch Group C). Of course, a more elegant explanation may present itself, a possibility to which we remain open in future work.

4. Implications and outlook

The overall goal of this paper was not to propose a self-standing alternative to Phase Theory or the PIC. Rather, it was to raise some rather undiscussed empirical and conceptual problems with it, and also to explore the kind of work that would need to be done to overcome those problems. To this end, we suggested that how competing instructions to transfer and label a phase are ordered may allow us to derive the attested variability in edge opacity. An interesting side effect of thinking about escape hatches as an interactive effect of Transfer and Labelling is that we have a relatively straightforward explanation for why edges appear to be transparent in traditional phase effects: it is essentially the next best option when Transfer is prioritized over Labelling. Put differently, escape hatches are simply a way of avoiding islands by manipulating instruction ordering in the grammar.

The reader may object that the problems surveyed here do not warrant the kind of revision we have proposed, but instead demand, for example, further work to preserve the restrictiveness of the PIC, or the abandonment of phases altogether. We are comfortable with either reaction, because, in our view, the present tenuous consensus about phases is

[7] We assume that subextraction is enforced by other factors, viz. information-structural considerations.

likely just a stepping stone on the way to a deeper understanding of locality phenomena generally. We hope to have contributed to that enterprise in a small way in this discussion.

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